232/3

PHYSICS

PAPER 3

MARKING SCHEME

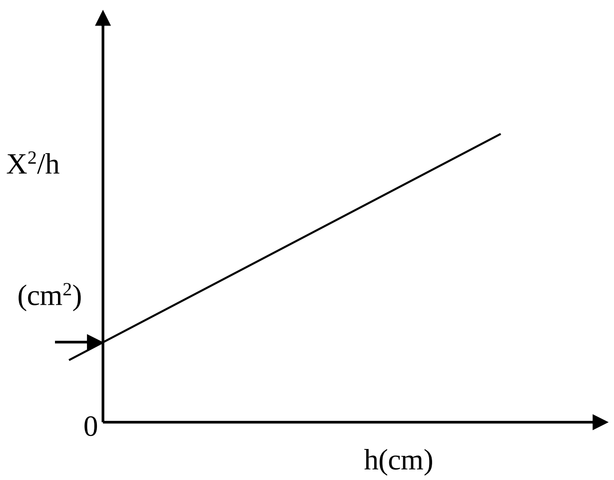
Q1. h= 5.0cm ± 1.0

Xcm	Hcm	X^2cm(cm^2)	X^2/h(cm)
20.0	5.0	400	400
25.0	7.0	625	89.29
30.0	10.0	900	90.00
35.0	13.0	1225	94.23
40.0	18.0	1600	88.89
45.0	24.0	2025	84.38

± 1.0

Use students own values.

Vi)



Axes-label/units -1
Scale -1
Plotting -2
Best line -1
5

Vii) Shown on graph -by student graph.

Xi) 28sec for 20 oscillations (student value within ± 1seconds).

X) $T = \frac{28}{20} = 1.4 \text{secs.}$

Xi) Making P subject and correct substitution best value of student.

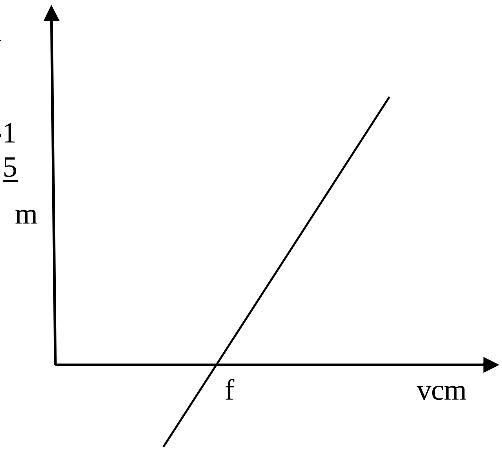
Q2. F = 20 ± 0.1.

e)

U	30	35	40	45	50	55
V	58	47.0	42.0	46	34.5	32.5
$M = {}^{v}/_{u}$	1.93	1.34	1.05	1.02	0.69	0.59

F) Plot.

-Axis -1
Scale -1
Plot -2
Curve -1
<u>5</u>



g) Slope = $\frac{1}{f}$

h) Use the x intercept
When M=0, v = f.

Or reciprocal of slope =f.